

4. What is Social Engineering?

Social engineering is the process of collecting credential information by means gaining their trust and further exploiting.

An example of

A common example of social engineering is phishing, where an attacker sends a fake email that looks like it is from a trusted company to steal login credentials.

1. Cyber security is the practice of protecting computer systems, networks, softwares and preventing any kind of illegal or malicious activities or access.

2. CIA stands for Confidentiality, Integrity and Availability.

Which refers that the credentials or personal data is secured and the data is maintained and available to the respective entity.

Confidentiality : Confidentiality ensures that information is only accessible to authorized individuals and is kept private from unauthorized access.

Integrity : Integrity ensures that data remains accurate, complete and unaltered except by authorized persons. It protects data from unauthorized modification or tampering.

Availability : Availability ensures that information and resources are accessible to authorized users when needed. Systems and data should be available and operational whenever required.

3. There are two types of hackers :

White Hat Hackers : Hackers who access or penetrate systems legally solely for the purpose of identification and fixing vulnerabilities so that they can be secured.

Black Hat Hackers : Hackers who penetrate systems for their own gain or malicious intentions. They exploit vulnerabilities to steal, damage or misuse data.

5. Ransomware is a type of malware which is a virus that encrypts a file and then demands a ransom or money to get access or to decrypt the files.

9. What is DoS attack?

A Denial of Service (DoS) attack is a type of cyber attack in which an attacker tries to make a computer, server, network or website unavailable to legitimate users by overwhelming it with a large amount of traffic or requests.

The main goal of a DoS attack is to exhaust the resources of the target system such as bandwidth, CPU, memory or connection limit so that it cannot respond to genuine users.

How it works :

- (i) The attacker identifies the target system.
- (ii) They send a huge number of requests or packets from one or more sources.
- (iii) The target system gets overloaded and is unable to handle normal traffic.
- (iv) Legitimate users are denied access to the service.

Types of DoS attack :

- (i) Volumetric attacks - Flood the target with large amount of traffic (e.g., UDP flood, ICMP flood).
- (ii) Protocol attacks - Exploit weaknesses in protocols to consume system resources (e.g., SYN flood).
- (iii) Application layer attacks - Target specific applications to exhaust resources (e.g., HTTP flood).

Prevention and Protection :

Use firewalls, intrusion detection systems (IDS), rate limiting, traffic filtering, and keep systems updated and well configured to reduce the impact of DoS attacks.